Block 2 — Intelligence · Creativity · Measurement · Problem Solving

Unit 1: Theories of Intelligence

Binet & IQ

Binet-Simon Scale (1905): 30 items, 50 children, French govt commission. Clinical interview, age-graded norms, mental age. **IQ = (MA/CA) × 100** (Stern, 1912/14). Stanford-Binet (Terman, 1916). 5 components: abstract reasoning, comprehension, clear direction of thought, purposeful thinking, self-corrective judgment.

Spearman's Two-Factor Theory (1904)

g = general intellective factor / "mental energy" (present in every mental act). **s** = specific factors. Group factors: verbal, numerical, mental speed. Method: factor analysis; **tetrad difference** ($F = r_{13}r_{24} - r_{14}r_{23} \approx 0$). Confirmed: Jensen (1998), Carroll (1993, Stratum III).
Critics: Thorndike (1926) — 3 intelligences (social/concrete/abstract) · Thomson (1939) — sampling theory · Thurstone (1935) — 7 primary abilities (later: hierarchical with g at top). 5 quantitative principles: conative control, fatigue, mental energy, primordial potencies, retentivity.

PASS Theory (Das, Nagliery & Kirby, 1994)

Grounded in **Luria (1966, 1973, 1980)**. 4 systems:
- P **Planning** — prefrontal cortex; programming, regulation, verification
- A **Attention-Arousal** — brain stem + reticular activating system
- S **Simultaneous** — occipital-parietal; *surveyability*
- S **Successive** — fronto-temporal; *serial order* (syntax, articulating speech)

+ **Knowledge base** (cultural/social). CAS = psychometric instrument.
Luria's 3 functional units: (1) arousal/attention (brain stem) (2) simultaneous/successive (occipital-parietal/fronto-temporal) (3) executive (prefrontal lobes)

Unit 2: Multiple Theories

Guilford's SOI (1956; modified 1977)

Rejected "g." 3 dimensions: **Contents** (5: visual, auditory, symbolic, semantic, behavioural) × **Operations** (5: cognition, memory, divergent production, convergent production, evaluation) × **Products** (6: units, classes, relations, systems, transformations, implications) = **150 cells** (originally 120). Cube representation. Open system.
Criticisms (Eysenck, 1972): restricted range · orthogonal rotation · factors too narrow.

Gardner's Multiple Intelligences (1983)

Intelligence = "bio-psychological potential to process information activated in a cultural setting." **8 criteria:** brain isolation, evolutionary history, core operations, encoding, developmental progression, idiot-savants/prodigies, experimental psychology, psychometrics.
8 intelligences: Linguistic · Logical-Mathematical · Musical · **Bodily-Kinesthetic** (use body to solve problems; learn by doing) · Spatial · Interpersonal · Intrapersonal · Naturalistic (1999). Existential = 9th candidate.
Criticisms: Morgan (1996) — cognitive styles · Hunt (2001) — not measurable · Eysenck (1990) — different abilities.

Sternberg's Triarchic Theory (1985, Beyond IQ)

3 subtheories:
Analytical **Componential:** metacomponents + performance + knowledge-acquisition (internal world)
Creative **Experiential:** novelty + automation (internal ↔ external)
Practical **Contextual:** adaptation + shaping + selection = "street smarts" (external world)
Successful intelligence = balance. Central feature = **adaptability**.
Criticisms: Gottfredson (2003) — unempirical · Das (2004) — hard to psychometrically measure.

Unit 3: Measurement of Intelligence

General Factor Theories (Tests)

Theory	Theorist	Key Idea
Two-Factor	Spearman (1904)	g + s
Primary Abilities	Thurstone (1935)	7 primaries (later: g at top)
SOI	Guilford (1956)	150 cells, no g
Multiple Intelligences	Gardner (1983)	8 (9) intelligences
Triarchic	Sternberg (1985)	Analytical/Creative/Practical
PASS	Das et al. (1994)	Planning/Attention/Simultaneous/Successive

Intelligence Tests

Binet-Simon (1905) → Stanford-Binet (Terman, 1916). **WAIS-IV** (Wechsler): verbal comprehension, perceptual reasoning, working memory, processing speed. **SB5:** fluid reasoning, knowledge, quantitative reasoning, visual-spatial processing, working memory. **KABC-II:** simultaneous + successive processing + planning + learning.

Unit 4: Creativity & Problem Solving

Creativity: Meaning

"Appropriate novelty" (Hennessey & Amabile, 1988). "Goal-directed thinking, unusual, novel, useful." **Newell, Shaw & Simon (1963)** — 4 criteria: novelty + usefulness; rejects accepted ideas; intense motivation; coherent new organisation.
Smith, Ward & Finke (1995): imaginative, purposeful, original, has value. **little-c** (everyday) vs **Big-C** (eminent) — Beghetto & Kaufman (2007).

Aspects of Creativity

Guilford (1986): divergent thinking: fluency, flexibility, originality, elaboration, sensitivity to problems, redefinition, freedom from functional fixedness.
Torrance (1966, 1974) TTCT: fluency · flexibility · originality · elaboration · abstractness of titles · resistance to premature closure.

Stages of Creativity — Wallas (1926)

Preparation (gather info, trial-and-error) → **Incubation** (unconscious processing) → **Illumination** (sudden insight) → **Verification** (test/check).

Investment & Confluence Theory (Sternberg, 2006)

6 resources: **Intellectual abilities** (synthetic + analytic + practical-contextual) · **Knowledge** (can help or hinder) · **Thinking styles** (legislative) · **Personality** (risk-taking, ambiguity tolerance, self-efficacy, overcome obstacles) · **Motivation** (intrinsic, task-focused; Amabile, 1983) · **Environment**.
"Buy low and sell high." Confluence: thresholds · partial compensation · multiplicative interaction.

Creativity & Intelligence

Terman (1925): 1,528 gifted (IQ 151) — none noteworthy creativity. Torrance (1977): r = .20 (178 correlations). **Threshold hypothesis** (Guilford, 1967): IQ ≥ 120 necessary not sufficient. **Triangularity hypothesis. Certification hypothesis** (Hayes, 1989): formal education required.

Measurement of Creativity

TTCT (Torrance). **Remote Associates Test (RAT)** (Mednick, 1962). **Wallach & Kogan (1965):** battery of 5 tests. Creativity ≠ IQ tests.